SHIVAM AI TRADER

Complete Setup Guide, A to Z: publish on GitHub, run on Google Cloud, or run on your own PC

This guide covers every way to run the bot, from creating the accounts to keeping it running. Read Part A first to choose your setup, then follow only the parts it points you to. Version of 7 November 2026 (bot with 84 automated tests).

Part	What it covers
A	Choose how to run the bot (4 options, and which combinations are allowed)
B	Create the accounts and keys: Telegram, Angel One, Gemini
C	Publish the bot on GitHub (web upload or Git), secrets, self-test, updates
D	Run on your own Windows PC (also Mac/Linux)
E	Google Cloud Run — instant replies, free tier (recommended)
F	Google Cloud virtual machine (VM) — everything always on
G	Settings (config.json)
H	Troubleshooting, security, maintenance and a quick command card

Important
Keep every token, key, PIN and secret private. They go only into GitHub **Secrets**, Google Cloud Run settings (the deploy script asks for them), or a private **.env** file on your own machine. Never paste them in a chat, a screenshot or a file you upload anywhere.

This bot is a learning and paper-trading tool. It never places orders and nothing it sends is financial advice. Read the companion PDF, *Features A to Z*, before relying on any signal.

Part A — Choose how to run the bot

The bot does two jobs: **answering your messages** (commands and button taps) and **sending automatic alerts** (crypto every 4 hours, stocks after the close, F&O snapshots). Each job can run in one of these places:

Option	Answers messages	Sends alerts	Cost	Your PC must be on?
1. GitHub Actions	Yes, every 5 min (often slower)	Yes, on schedule	Free (public repo)	No
2. Google Cloud Run	Yes, in seconds	No (GitHub does it)	Free tier (card needed)	No
3. Google Cloud VM	Yes, in seconds	Yes	≈ $3–4/month for the IP address	No
4. Your own PC	Yes, in seconds	Yes	Free (electricity)	Yes, awake and online

Recommended setups

Setup	Who does what	Follow parts
Best (recommended)	GitHub sends alerts · Cloud Run answers instantly	B, C, E
Simplest	GitHub does everything (slow replies)	B, C
Own PC	Your PC does everything while it is on	B, D (C optional, for backup/updates)
Own PC + GitHub alerts	PC answers (local --no-alerts) · GitHub sends alerts	B, C, D
Cloud VM	VM does everything, 24×7	B, C (to store the code), F

Important

Two rules that prevent double or missing replies:

1. **Only one place answers messages.** Telegram can deliver messages either to a webhook (Cloud Run) or to polling (GitHub, PC, VM), never both. Cloud Run sets a webhook; to answer from a PC or VM, first run `python -m bot.main webhook delete`, and disable GitHub's **Bot commands** workflow.
2. **Only one place sends alerts.** If your PC or VM runs `bot.main local` (with alerts), disable GitHub's **Market alerts** workflow, or use `local --no-alerts`.

Part B — Accounts and keys

B1. Telegram bot token and chat ID (required)

1. Open Telegram, search **@BotFather** (blue tick), send **/newbot**.
2. Choose a name (e.g. *Shivam Trader*) and a username ending in **bot**.
3. Copy the **token** BotFather sends (123456789:AAH…). This is **TELEGRAM_BOT_TOKEN**.
4. Open your new bot, tap **Start** and send any message, e.g. *hi*.
5. In a browser open the address below (put your token in place of <TOKEN>).

```
https://api.telegram.org/bot<TOKEN>/getUpdates
```

Find `"chat":{"id":` and copy the number after it. This is **TELEGRAM_CHAT_ID**. The bot answers only this chat. If the page shows `"result":[]`, send another message and refresh. If Cloud Run is already connected, this page shows an error instead; your chat ID is unchanged, so use the one you

saved.

B2. Angel One SmartAPI (for live NSE, F&O and futures)

Without it, Indian stocks use delayed Yahoo data and F&O, futures and OI build-up are off. Crypto, fundamentals and news work either way.

1. Open the official **Angel One SmartAPI** developer portal and sign in with your Angel One client ID.
2. **Create an app** and copy its **API key** → **ANGEL_API_KEY**.
3. Find **Enable TOTP** (portal or Angel One app). Copy the **text secret** shown with the QR code → **ANGEL_TOTP_SECRET**.
4. Your Angel One client ID → **ANGEL_CLIENT_CODE**; your login PIN → **ANGEL_PIN**.

> **Risk**
> Together these four values can log in to your Angel One account. The bot only reads data and has no code that places, changes or cancels orders. Keep them private; if you suspect a leak, regenerate the API key and TOTP in Angel One at once.

B3. Gemini key (optional, for AI summaries)

1. Go to **aistudio.google.com** → **Get API key** → **Create API key** → **GEMINI_API_KEY**.
2. Open a current **Flash** model in AI Studio, click **Get code** and copy the exact model ID from the code → **GEMINI_MODEL**.

> **Note**
> If the self-test shows "Gemini failed … HTTP 404", the model ID is wrong or retired: copy a newer one from **Get code**. The bot works fine without Gemini.

Part C — Publish the bot on GitHub

C1. Account and safety

1. Sign up at **github.com**.
2. Turn on two-factor login: profile picture → **Settings** → **Password and authentication** → **Enable two-factor authentication**.

C2. Create the repository

1. Click **+** (top right) → **New repository**.
2. Name: `shivam-ai-trader-bot`. Choose **Public**. Don't add a README. Click **Create repository**.

> **Note**
> **Why public?** Public repositories get free GitHub Actions minutes. Others can see the code and run times, never your secrets or Telegram messages. If you choose private, change `*/5` to `*/30` in `.github/workflows/commands.yml` to stay inside the free minutes.

C3. Upload the files — way 1: in the browser (easiest)

1. On a computer, unzip **shivam-ai-trader-bot.zip** and open the **shivam-ai-trader-bot** folder.
2. On the new repository page click **uploading an existing file**.
3. Select **everything inside** the folder (Ctrl+A) and drag it onto the page (Chrome or Edge).
4. Click **Commit changes**.
5. Check that the repository shows `bot/`, `tests/` and `.github/workflows/` with 3 files. If `.github` is missing, drag that folder in again.

> **Important**
> Never upload a `.env` file or a `.venv` folder if you created them while running the bot on your PC (Part D). The `.env` file holds your secrets.

C4. Upload the files — way 2: with Git (better for frequent updates)

Install **Git for Windows** from git-scm.com (default options). Then open PowerShell in the unzipped `shivam-ai-trader-bot` folder (Shift + right-click inside the folder → **Open PowerShell window here** or **Open in Terminal**) and run, using your GitHub username:

```
git init
git config user.name "Your Name"
git config user.email "you@example.com"
git add .
git commit -m "SHIVAM AI TRADER bot"
git branch -M main
git remote add origin https://github.com/YOUR-USERNAME/shivam-ai-trader-bot.git
git push -u origin main
```

A browser window asks you to sign in to GitHub the first time. Git skips `.env` and `.venv` automatically because they are listed in `.gitignore`.

Later updates with Git: copy the new files over the old ones, then:

```
git add .
git commit -m "Update"
git push
```

C5. Add the secrets

Repository → **Settings** → **Secrets and variables** → **Actions**. Names must match exactly.

Secrets tab → New repository secret	Value	Needed?
TELEGRAM_BOT_TOKEN	Token (B1)	Yes
TELEGRAM_CHAT_ID	Chat ID (B1)	Yes
ANGEL_API_KEY	SmartAPI key (B2)	For NSE live, F&O, futures
ANGEL_CLIENT_CODE	Client ID (B2)	Same
ANGEL_PIN	Login PIN (B2)	Same
ANGEL_TOTP_SECRET	TOTP text secret (B2)	Same
GEMINI_API_KEY	Gemini key (B3)	Optional

Variables tab → New repository variable: `GEMINI_MODEL` = the model ID (B3).

C6. Turn on and test

1. Open the **Actions** tab. If GitHub asks, click the button to enable workflows.
2. You should see **Market alerts**, **Bot commands** and **Self-test**.
3. Click **Self-test** → **Run workflow** → **Run workflow**.
4. Within about 2 minutes Telegram receives a self-test message. Best on a weekday, 09:15–15:30 IST, so Angel One Greeks and order book are available.
5. Send `/start` to your bot. Without Cloud Run, the reply comes within 5–10 minutes.

Self-test line	What it checks
Crypto data / order book	Binance public data (Kraken as fallback)
Stock data	NSE candles from Angel One (or Yahoo)
Angel One login, instruments, NIFTY, OI, Greeks	Your four ANGEL_ secrets and the option chain
Futures	All NIFTY expiries, OI history, OI build-up
News feeds	How many public news feeds answered
Fundamentals	Yahoo company data for the first watchlist stock
Gemini	Your key and model

> **Tip**
> If any line shows ⚠ or ✗, forward that Telegram message to Claude. It lists the exact field names Angel One returned, so the code can be adjusted.

C7. Faster replies without Cloud Run

GitHub app or website → **Actions** → **Bot commands** → **Run workflow** → type a command such as `/stock TATAMOTORS` → **Run**. The answer arrives in 1–2 minutes.

C8. Updating the bot on GitHub later

- Browser: **Add file → Upload files**, drag in the changed files keeping the same folders, **Commit changes**. Secrets stay as they are.
- Git: copy files, then `git add . ; git commit -m "Update" ; git push`.
- After updating, also update Cloud Run (E7) or your PC/VM (D8, F5) if you use them.

> **Note**
> **60-day rule:** GitHub pauses scheduled workflows in public repositories after 60 days with no changes, and emails you first. Click **Enable workflow** in the Actions tab, or edit any file to reset the timer.

Part D — Run on your own Windows PC

Your PC answers messages in seconds and sends the scheduled alerts, as long as it is on, awake and online. No GPU is needed. Everything runs in a private Python environment inside the bot folder.

D1. Install Python 3.12

1. Go to **python.org** → **Downloads** → **Windows** → download the latest **Python 3.12** 64-bit installer.
2. Run it and **tick "Add python.exe to PATH"** on the first screen, then **Install Now**.
3. Open PowerShell and check: `py -3.12 --version` (shows Python 3.12.x).

D2. Get the bot onto the PC

- **From the zip:** unzip `shivam-ai-trader-bot.zip` to a simple path, e.g. `C:\Bots\shivam-ai-trader-bot`.
- **From GitHub (with Git installed):** `git clone https://github.com/YOUR-USERNAME/shivam-ai-trader-bot.git`

D3. Easy way: double-click start_bot.bat

1. Open the bot folder and double-click **start_bot.bat**.
2. The first time, it creates the Python environment (`.venv`) and installs everything (5–10 minutes). Then it copies `.env.example` to `.env` and opens it in Notepad.
3. Fill in your values (Part B), e.g. `TELEGRAM_BOT_TOKEN=123456789:AAH…`, save, close Notepad.
4. Double-click **start_bot.bat** again. The window shows *SHIVAM AI TRADER is running with alerts*. Send `/start` to your bot.
5. To stop: press **Ctrl+C** in the window, or close it.

> **Important**
> If Windows shows "Windows protected your PC", click **More info → Run anyway** (the file is a plain text script you can open in Notepad to read). If the window says a webhook is connected, see D6.

D4. Manual way (PowerShell)

Open PowerShell in the bot folder and run these one by one:

```
py -3.12 -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
pip install -r requirements.txt
copy .env.example .env
notepad .env
```

If `Activate.ps1` is blocked ("running scripts is disabled"), run this once and answer **Y**, then activate again:

```
Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
```

D5. Test and run

```
python -m pytest -q            # 84 offline tests, about 15 seconds
python -m bot.main selftest    # live check, result arrives in Telegram
python -m bot.main local       # answer messages + send alerts (Ctrl+C to stop)
python -m bot.main local --no-alerts   # answer messages only (GitHub keeps alerts)
python -m bot.main run "/stock INFY"   # run one command once
```

Next time you only need: open PowerShell in the folder → `.\.venv\Scripts\Activate.ps1` → `python -m bot.main local` (or just double-click start_bot.bat).

D6. Avoid conflicts

If you also use…	Do this
Google Cloud Run (webhook)	Run `python -m bot.main webhook delete` before starting the PC. To go back to Cloud Run later, rerun `bash deploy_cloudrun.sh` in Cloud Shell.
GitHub Bot commands workflow	Actions → Bot commands → ⋯ → **Disable workflow** while the PC answers (enable it again when the PC is off for long).
GitHub Market alerts workflow	Disable it if the PC runs `local` with alerts, or run the PC with `--no-alerts`.

D7. Keep it running

- **Stop sleep:** Settings → System → Power → **Screen and sleep** → "When plugged in, put my device to sleep after" = **Never**.
- **Start automatically:** press Win+R → `taskschd.msc` → **Create Basic Task** → name it *Shivam bot* → trigger **When I log on** → action **Start a program** → browse to **start_bot.bat** → Finish.
- Alerts that fall due while the PC is off or asleep are skipped (a scan runs if the PC is on within 30 minutes of its time). Times match GitHub: crypto ≈01:37, 05:37, 09:37, 13:37, 17:37, 21:37 IST; stocks 16:10; F&O 09:45 and 14:30 IST on weekdays.

D8. Update the bot on your PC

- Zip: replace the files with the new ones (keep your `.env` and `.venv`), then run `pip install -r requirements.txt` inside the activated environment.
- Git: `git pull`, then the same `pip install -r requirements.txt`.

D9. Mac or Linux

```
cd shivam-ai-trader-bot
chmod +x start_bot.sh
./start_bot.sh      # first run sets up .venv and creates .env
nano .env           # fill in your values, Ctrl+O Enter Ctrl+X
./start_bot.sh      # runs the bot (Ctrl+C to stop)
```

Part E — Google Cloud Run (instant replies, recommended)

Telegram sends each message straight to a small server that Google starts on demand, so replies arrive in seconds and nothing runs while you are quiet. GitHub keeps sending the alerts.

E1. Account, project and budget

1. Go to **console.cloud.google.com** and sign in with Google. Accept the terms and start the free trial (a card is needed for verification; new accounts get trial credit).
2. Top bar → project picker → **New project** → name `shivam-trader` → **Create**, then select it.
3. Menu → **Billing** → **Budgets & alerts** → **Create budget** → amount e.g. ₹100 → alerts at 50%, 90%, 100% → **Finish**. You get an email if anything starts costing money.
4. When the trial ends, Google asks you to **upgrade** the billing account to keep the service running. The always-free limits still apply after upgrading.

E2. Open Cloud Shell and get the code

1. Click the **Cloud Shell** icon >_ (top right). A terminal opens at the bottom.
2. Make sure it uses your project: `gcloud config set project YOUR_PROJECT_ID` (the ID is shown in the project picker).
3. Download your repository from GitHub (Part C):

```
git clone https://github.com/YOUR-USERNAME/shivam-ai-trader-bot.git
cd shivam-ai-trader-bot
```

E3. Deploy

```
bash deploy_cloudrun.sh
```

Type each value when asked (secrets are hidden while you type) and press Enter. The script:

- turns on Cloud Run, Cloud Build and Artifact Registry (first time takes a minute);
- builds and deploys the bot (3–6 minutes the first time) with 1 GB memory, up to 3 servers, zero when idle;
- deletes old container images, keeping the 2 newest and the one in use, and sets an automatic cleanup policy (images older than 1 day are deleted daily) so storage stays under the free 500 MB; it prints how much is used;
- sets uploaded source files to auto-delete after 1 day;
- connects Telegram to the service with a secret, one message at a time.

E4. Finish

1. On GitHub: **Actions → Bot commands → ⋯ → Disable workflow** (Cloud Run answers now). Keep **Market alerts** and **Self-test** on.
2. Send `/start`. The first reply after a quiet period can take 10–20 seconds while the server wakes up; after that, a few seconds.

E5. Everyday commands in Cloud Shell

Task	Command (inside the shivam-ai-trader-bot folder)
Update after new code on GitHub	`git pull && bash deploy_cloudrun.sh` (press Enter to keep each value)
Clean old images only	`bash deploy_cloudrun.sh cleanup`
Stop instant replies	`bash deploy_cloudrun.sh disconnect`, then re-enable GitHub's Bot commands workflow
Change one secret	Run the deploy again and type only that value; Enter keeps the others

E6. Where to look

- **Logs:** Menu → **Cloud Run** → `shivam-trader-bot` → **Logs**. Errors from commands appear here.
- **Storage:** Menu → **Artifact Registry** → `cloud-run-source-deploy`.
- **Costs:** Menu → **Billing → Reports**.

Free each month (always-free tier)	Normal use of this bot
Cloud Run: 2 million requests, 180,000 vCPU-seconds, 360,000 GiB-seconds	A few hundred messages; well inside
Artifact Registry: 0.5 GB storage	2–3 images (the script keeps it there)
Cloud Build: free build minutes each month	A few minutes per deploy

> **Note**
> Free-tier limits are set by Google and can change. The budget alert (E1) is your safety net. Choose region us-central1 (the script's default) for the free tier.

E7. Update order (after Claude sends new files)

1. Upload the changed files to GitHub (C8).
2. Cloud Shell: `cd shivam-ai-trader-bot && git pull && bash deploy_cloudrun.sh`, pressing Enter at every question.
3. Run the Self-test on GitHub (C6) to confirm.

Part F — Google Cloud virtual machine (everything always on)

A small Linux computer in Google's cloud runs the bot 24×7, exactly like your own PC (Part D) but without keeping your PC on. Use it instead of Cloud Run if you want alerts and replies from one place.

> **Cost**
> **Not completely free.** One e2-micro VM in us-west1, us-central1 or us-east1 with a 30 GB **standard** disk is in the always-free tier, but its public IPv4 address is charged (about $0.005 per hour, ≈ $3.65 a month at the time of writing). The VM also has only 1 GB memory, so the steps add swap space. Cloud Run (Part E) is cheaper.

F1. Create the VM

1. Console → Menu → **Compute Engine → VM instances** → enable the API if asked → **Create instance**.
2. Name `shivam-bot` · Region **us-central1** · Machine type **e2-micro**.
3. **Boot disk → Change**: Debian 12, disk type **Standard persistent disk**, 30 GB → Select.
4. Leave firewall boxes unticked (the bot only makes outgoing connections). Click **Create**.

F2. Install the bot

Click **SSH** next to the VM. In the window that opens, run:

```
sudo apt update && sudo apt install -y python3-venv python3-pip git
sudo fallocate -l 2G /swapfile && sudo chmod 600 /swapfile
sudo mkswap /swapfile && sudo swapon /swapfile
echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab
git clone https://github.com/YOUR-USERNAME/shivam-ai-trader-bot.git
cd shivam-ai-trader-bot
./start_bot.sh          # first run: sets up .venv, creates .env, stops
nano .env               # fill in values; save with Ctrl+O, Enter, Ctrl+X
.venv/bin/python -m bot.main selftest
```

F3. Run it as a service (starts on boot, restarts on errors)

```
sudo tee /etc/systemd/system/shivam-bot.service >/dev/null <<EOF
[Unit]
Description=SHIVAM AI TRADER bot
After=network-online.target
Wants=network-online.target

[Service]
User=$USER
WorkingDirectory=$HOME/shivam-ai-trader-bot
ExecStart=$HOME/shivam-ai-trader-bot/.venv/bin/python -m bot.main local
Restart=always
RestartSec=10

[Install]
WantedBy=multi-user.target
EOF
sudo systemctl daemon-reload
sudo systemctl enable --now shivam-bot
sudo systemctl status shivam-bot      # should say active (running)
journalctl -u shivam-bot -f           # live log, Ctrl+C to leave
```

F4. Avoid conflicts

- If Cloud Run was connected: `.venv/bin/python -m bot.main webhook delete` (the service refuses to start while a webhook is set).
- On GitHub disable **Bot commands** and **Market alerts** (the VM does both). Keep **Self-test**.
- Or add `--no-alerts` to the ExecStart line to leave alerts on GitHub.

F5. Update, stop, delete

```
cd ~/shivam-ai-trader-bot && git pull
.venv/bin/pip install -r requirements.txt
sudo systemctl restart shivam-bot
```

Stopping the VM (VM instances → ⋯ → Stop) stops the bot; a stopped VM still pays for a reserved static IP if you created one. **Delete** the VM to stop all charges.

Part G — Settings (config.json)

Edit `config.json` on GitHub (pencil icon → Commit changes) or on your PC/VM. Cloud Run needs a redeploy (E5) to pick up changes; a PC/VM needs a restart.

Setting	Default	Meaning
risk.capital_inr	15000	Capital for NSE stocks/F&O (position size, risk per lot)
risk.capital_usdt	100	Capital for crypto
risk.risk_per_trade_pct	0.5	Maximum loss per trade, % of capital
risk.target_r	2.0	Take profit = this many times the risk
crypto.timeframe	4h	Crypto candle size
crypto.top_n_by_volume	15	Most-traded coins scanned by /crypto
crypto.always_include	BTC, ETH, SOL, BNB, XRP	Coins always scanned
stocks.watchlist	12 large NSE stocks	Stocks scanned by /stocks and shown as quick buttons
stocks.indices	NIFTY, BANKNIFTY	Indices added to /stocks with Angel One
fno.strikes_each_side	10	Option strikes fetched above and below the price
fno.auto_snapshot	NIFTY, BANKNIFTY	Automatic F&O snapshots
alerts.max_per_run	5	Maximum alerts per scan
costs.*	0.10% / 0.15% / 0.05%	Fees per side (crypto/stocks) and slippage in backtests

> **Important**
> Keep JSON punctuation exact: names and text in double quotes, commas between items, no comma after the last item. A typo stops the bot until it's fixed.

Part H — Troubleshooting, security and quick reference

Problem	Cause and fix
CI "test" job fails: No module named 'flask'	Old requirements.txt. Upload the latest one (it lists flask and gunicorn).
No reply on GitHub-only setup	Bot commands runs every 5 min but GitHub can be late (up to an hour after first upload). Use Run workflow (C7) or Cloud Run.
No reply with Cloud Run	Check Cloud Run logs (E6). Re-run the deploy and type the token so the webhook is set again.
PC/VM says a webhook is connected	`python -m bot.main webhook delete`, then start again.
Replies twice / alerts twice	Two places are answering or alerting. Follow the two rules in Part A.
"Angel One login failed"	Wrong PIN or TOTP text secret, or TOTP not enabled.
"NSE stock … not found"	Use the NSE symbol, or type part of the name and pick from the buttons (e.g. `tata`).
Futures/F&O "needs Angel One"	Add the four ANGEL_ values (GitHub secrets, Cloud Run deploy, or .env).
Greeks or order book empty	Normal outside 09:15–15:30 IST on weekdays.
Fundamentals "no data"	Yahoo has no data for that symbol or blocked the server; try later.
News "No fresh stories"	Feeds unreachable at that moment; tap Refresh later. Self-test shows how many feeds work.
Gemini failed (HTTP 404/400)	Copy the exact model ID from AI Studio's Get code (B3).
pip install fails on Windows	Use Python 3.12 (not a preview version); run `python -m pip install --upgrade pip` first.
Storage above 0.5 GB on Google Cloud	`bash deploy_cloudrun.sh cleanup` in Cloud Shell.

Security checklist

- Two-factor login on GitHub and Google.
- Secrets only in GitHub Secrets, the Cloud Run deploy prompts, or `.env`; `.env` never uploaded (it is in .gitignore and .gcloudignore).
- The bot answers only your TELEGRAM_CHAT_ID; the Cloud Run endpoint rejects calls without Telegram's secret header.
- If anything leaks: Telegram BotFather → /revoke; regenerate Angel One API key and TOTP; delete and recreate the Gemini key; update every place that stores them.
- The bot never places orders. SEBI's 2026 retail algo rules (static IP, Algo ID, broker approval) apply to API *orders*, not to reading data.

Quick command card

Where	Command
Telegram	/start · /crypto · /stocks · /stock SYMBOL · /search name · /fundamentals SYMBOL · /fno SYMBOL · /futures SYMBOL · /buildup · /news [india|world|crypto|SYMBOL] · /strategies · /settings · /help · a number from the last list
PC / VM	`python -m bot.main local` · `local --no-alerts` · `selftest` · `run "/stock INFY"` · `webhook delete` · `alerts crypto|stocks|fno`
Cloud Shell	`bash deploy_cloudrun.sh` · `… cleanup` · `… disconnect`
Tests	`python -m pytest -q` (84 tests)